factors or genes of one pair assort out independently of the other pair."

More to Know

T.H. Morgan was awarded Nobel Prize in 1993 for determining the role of chromosomes in heredity.

18.5 Chromosomes, DNA and Genes

The human body is made up of million cells. The nucleus of each cell contains thin thread like structures called **chromosomes**. The term 'chromosomes' was first coined by **Waldeyer** in 1888. The chromosomes are the carrier of genetic material which contain the heredity information.

The chromosomes are highly condensed coiled chromatin fibres packed with the **DNA** (Deoxyribonucleic acid) that forms the genetic material. **Genes** are segments of DNA, which are responsible for the inheritance of a particular phenotypic character. Each gene is present at a **specific position** on a chromosome called its **locus**. During cell division, the genetic information present in the genes are passed from one generation to another.

18.5.1 Structure of a Chromosome

The chromosomes are thin, long and thread like structures consisting of two identical strands called sister chromatids. They are held together by the centromere. Each **chromatid** is made up of spirally coiled thin structure called **chromonema.** The chromonema has number of bead-like structures along its length which are called **chromomeres.** The chromosomes are made up of DNA,RNA, chromosomal proteins (histones and non-histones) and certain metallic ions. These proteins provide structural support to the chromosome .

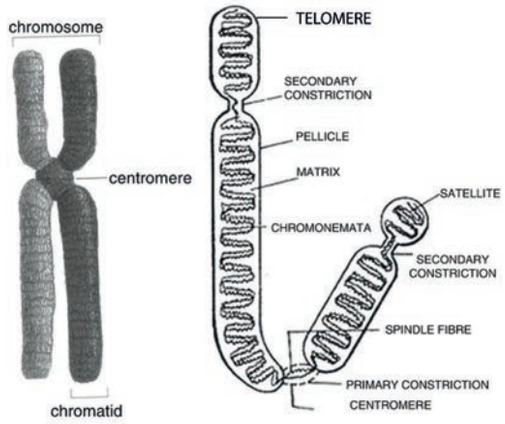


Figure 18.3 Structure of chromosome

A chromosome consists of the following regions

Primary constriction: The two arms of a chromosome meet at a point called **primary constriction** or **centromere**. The centromere is the region where spindle fibres attach to the chromosomes during cell division.

Secondary constriction: Some chromosomes possess secondary constriction **at any point** of the chromosome. They are known as the nuclear zone or **nucleolar organizer** (formation of nucleolus in the nucleus).

Telomere: The **end of the chromosome** is called telomere. Each extremity of the chromosome has a polarity and prevents it from joining the adjacent chromosome. It maintains and provides **stability to the chromosomes.**

Satellite: Some of the chromosomes have an elongated **knob-like appendage** at one end of the chromosome known as satellite. The chromosomes with satellites are called as the **sat-chromosomes**.

DO YOU KNOW?

Telomeres act as aging clock in every cell.

Telomeres are protective sequences of nucleotides found in chromosomes. As a cell divides every time, they become shorter. Telomeres get too short to do their job, causing our cells to age.

10th_Science Unit-18.indd 265 06-12-2021 18:45:05